ARE WE CHASING SHADOWS? QUANTIFYING UNATTRIBUTABLE POLARIZATION AND ATTRIBUTING THE REST TO ANNOTATOR GROUPS

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ABSTRACT

Polarization has recently been proposed as a robust way of distinguishing minor disagreements from systematic differences in opinion for imbalanced, minority groups but there currently exists no methodology for attributing polarization to these groups. In this study, we discover and investigate two major limitations in realistic settings: (1) the presence of "inherent" polarization that cannot be attributed to any known or latent groups, and (2) opposing polarization effects canceling each other out in aggregated annotations. To address these issues, we introduce a new metric that attributes polarization to any annotator group while avoiding these limitations, tailored to smaller datasets common in NLP, and provide an open-source Python library implementation. We apply our method to four content moderation datasets and confirm that gender and ethnicity consistently explain polarization patterns, while ideology and personal experiences may be more important than previously thought. Our work overall indicates that subjective NLP datasets may need fewer items and more annotators per item.

1 INTRODUCTION

Annotations are essential for subjective tasks, such as content moderation, toxicity, and Hate Speech (HS) detection, which are especially important for protecting minority groups United Nations (2025); Dioum (2025). However, in practice, analysis rests on inter-annotator agreement or majority-vote Ivey et al. (2025); Kralj Novak et al. (2022); van der Velden et al. (2025), which may lead to biased and misconfigured systems (e.g., toxicity models disproportionally marking African American (AA) speech as "toxic" Sap et al. (2019)), or discard *fundamental* disagreements between minority groups and the majority Quaranto (2022), (see the example in Fig. 1).[1]

One alternative is using *polarization*, which correlates better with human judgments and is much less sensitive to different group sizes Pavlopoulos & Likas (2023; 2024). While we can detect polarization, there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question (RQ):

RQ. *How much polarization can we practically attribute to annotator subgroups given existing datasets?*

While prior work assumed that all polarization can be explained and attributed to some annotator characteristic, through the exhaustive creation of theoretical annotator groups our study discovers the existence of Inherent Polarization (InPol), which can not be attributed to *any possible group* of annotators—known or unknown (Finding 1). We also discover that polarization across comments has a *direction*, preventing us from aggregating annotations on the dataset level to circumvent annotation sparsity (Finding 2).

In order to solve these issues we introduce a new metric—"*apunim*"—which attributes polarization to specific annotator sociodemographic and ideological characteristics (referred to as Personal Char-

[1] The simulated experiments in this study were implemented by generating random annotations following the normal distribution with differing means and Standard Deviation (SD)—see App. F.

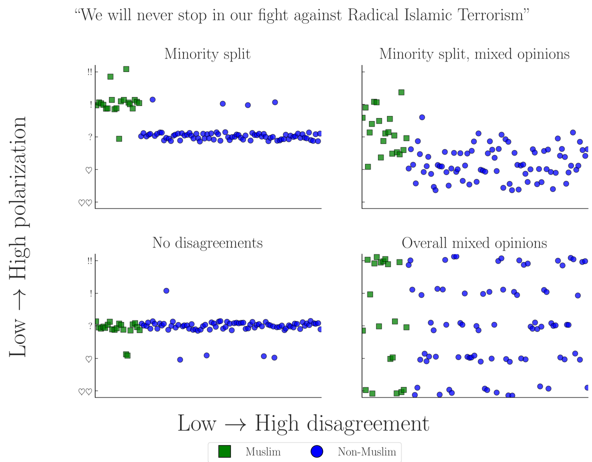

Figure 1: Simulated experiment exploring four scenarios of a HS detection task applied to a controversial statement by D. Trump (Jan. 2019). The variance in annotations across different annotators may cause traditional agreement metrics to ignore minority opinions, especially when opinions are mixed (top-right).

acteristics (PCs) in this study). Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables both *quantitative* comparisons across datasets and statistical analysis via p-value estimation. Even in cases where unattributable polarization exists, *apunim* is capable of detecting general patterns of polarization across multiple groups. Additionally, our metric is designed to be used for existing Natural Language Processing (NLP) datasets which are usually small w.r.t. both content and annotations due to budget constraints Chen et al. (2022); Golazizian et al. (2024).

We then apply our metric to four NLP datasets focused on Toxicity and HS detection (§6), finding that annotator gender and ethnicity significantly impact annotation (Finding 4). Our findings also indicate that ideological attributes and the annotators' pesonal experiences may play a major role in explaining polarization (e.g., the more religious the annotators, the more polarized they become—Finding 5), a possibility not adequately explored by relevant literature. Finally, we empirically investigate and discuss the applicability of our methodology to the modern NLP dataset landscape (App. J), and release an open source Python (PyPi) library that implements our metric.[2]

Warning: This publication includes examples of controversial and potentially harmful speech for the purposes of demonstration. These examples do not represent the views of the authors.

2 Background and related work

2.1 Agreement

Popular inter-annotator agreement metrics such as Cohen's kappa, Krippendorff's alpha, and Fleiss's kappa aim to distinguish genuine disagreement from random disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup agreement without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds (Feinstein & Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), producing results that are difficult to compare across studies, being sensitive to class imbalance (Feinstein & Cicchetti, 1990), and relying on questionable assumptions about chance agreement Checco et al. (2017). While recent work has addressed some of these limitations with methods that improve applicability and robustness van der Velden et al. (2025); Wong et al.

[2]Library: `https://anonymous.4open.science/r/apunim-5F70`, Webpage: `https://anonymous.4open.science/r/apunim-page`, Replication code: `https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`

Dataset	#Com	#Ann	#Dims	Task
Kumar	2998	5	10	Tox
Sap	585	5–6	3	HS
DICES-350	350	123	4	HS
DICES-990	990	70–76	4	HS

Table 1: Datasets used in this study. *#Ann* refers to the number of annotations per comment (see App. B), and *#Dims* to PC dimensions (e.g., age, gender). Tasks are either Toxicity (Tox) or HS.

(2021); Ivey et al. (2025), they are much more complex and less intuitive compared to their traditional counterparts, indicating that agreement may not be the correct tool for analysis of minority viewpoints.

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches Akhtar et al. (2019) utilizes χ^2 to estimate in-group vs out-group variance, but still requires traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric Checco et al. (2017), or non-parametric Pavlopoulos & Likas (2024); Mignemi et al. (2024) approaches to the annotation histograms.

For our work, we use a non-parametric polarization metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos & Likas, 2024). nDFU takes values in the range $[0, 1]$, where values close to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). The metric is defined as follows:

$$\mathrm{nDFU}(X) = \frac{\max\limits_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m} \tag{1}$$
$$m = \arg\max_i f_i$$

where, X denotes the set of annotations, f_i the relative frequency of the i-th annotation (e.g., the frequency of 2 in a scale of 1–5), and m the histogram peak. Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak (see Fig. 2 for an example).

2.3 Attributing polarization to groups

While many studies study the effect of subgroups on annotation agreement Goyal et al. (2022); Binns et al. (2017); Giorgi et al. (2025); Al Kuwatly et al. (2020); Sachdeva et al. (2022), attributing *polarization* is a relatively new concept. Under the name *aposteriori unimodality* Pavlopoulos & Likas (2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then it is explained by those groups. Though promising, this mechanism operates on the assumption that *all* polarization can be explained by a single split, which we disprove prove to be impossible in many cases (§4). Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.[3]

3 Datasets

We use four datasets from current NLP research that include PC information at the level of individual annotators; specifically, the “Kumar” Kumar et al. (2021) and “Sap” (“Breadth of Posts”) Sap et al.

[3]For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to $.001$, and cases where attribution is $.49$.

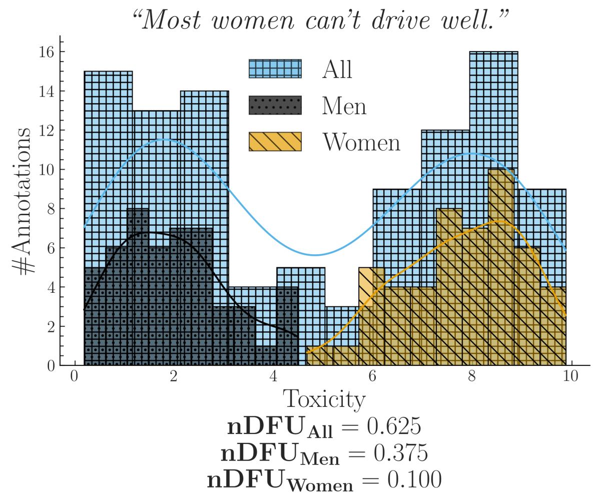

Figure 2: Simulated experiment describing a polarizing comment where male and female annotators agree between themselves, but disagree with the opposite gender.

(2022) datasets, which are concerned with Toxicity and HS detection respectively, and the DICES-350 and DICES-990 datasets Aroyo et al. (2023), which focus on LLM response safety.[4]

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct PC dimensions.[5] In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. More details on the annotations, including overall levels of observed polarization, can be found in App. B.

4 Is it possible to attribute all polarization to annotator groups?

4.1 Problem formulation

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated comments.[6] Each comment c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \ldots\}$, where a_i is a single annotation (e.g., 2 in a LIKERT scale of 1–5), and g_i a group (e.g., men). As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{“Could be better, could be worse”}) \\ = \{\ (+, F), (-, M), (+, F), \ldots\ \}$$

Intuitively, a group partially explains the polarization of a comment c when the annotations of that group $A(c \mid g) = \{a(c, g') \in A(c) \mid g' = g\}$ exhibit higher polarization compared to the full set of annotations $A(c)$. Consider the case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2. The annotations are generally polarized ($nDFU_{all} = .62$); but both female ($nDFU_{women} = .10$) and male ($nDFU_{men} = .37$) annotators exhibit low

[4]For the latter, we consider only labels relating to racism (`Q3_bias_overall`), thereby transforming the task to HS detection.

[5]We sample 3,000 comments in this study due to computational constraints (App. E). We run ablations for both different and differently sized-samples. Our results remain consistent, although large samples (10,000+) tend to produce null results (App. H).

[6]The annotated items could be of any modality, such images, videos or survey questions, even though we only consider comments in this study.

polarization between their respective groups. This pattern suggests that the overall polarization is partially caused by substantive disagreements between male and female annotators.[7]

4.2 Inherent polarization

4.2.1 What is it?

Inherent Polarization (InPol) refers to disagreement that cannot be attributed to *any specific* grouping of annotators; that is, polarization that exists no matter how we split the annotators in any arbitrary group. Prior work Pavlopoulos & Likas (2024) attributed polarization to annotator groups only when the observed polarization for each individual group became exactly zero (e.g. $nDFU(all) > nDFU(men) = nDFU(women) = 0$). The presence of InPol would invalidate this theory, since there would be no possible group or combination of groups g *s.t.* $nDFU(g) = 0$.

Definition 1 (Inherent Polarization)**.** Polarization that cannot be explained by any known or latent annotator grouping.

Formally, the InPol for a single comment c is the minimum polarization exhibited by any arbitrary group:

$$Pol_{inh}(c) = \min_{S \subseteq A(c),\, |S| \geq 3} \{\textit{nDFU}(S)\} \tag{2}$$

When considered exhaustively, these theoretical, random groups include both observed and latent groups (e.g., annotators may be less strict if they had lunch prior to annotation). Since nDFU operates only when a group has at least three annotators (Eq. 1), this yields $\sum_{k=3}^{n} \binom{n}{k}$ possible groups.

4.2.2 Experimental setup

We systematically explore the space of annotator groups by generating random partitions for each comment. Since enumerating all subsets is computationally intractable for large annotator counts, we establish an upper bound via Monte Carlo sampling by drawing k random partitions with random sizes:

$$Pol_{inh}^{MC}(c) = \min_{j=1}^{k} \{\textit{nDFU}(\tilde{r}_j(A(c)))\} \tag{3}$$

where $\tilde{r}$ is an operator that selects a randomly sized, randomly selected subset. Since Eq. 3 considers a strict subset of Eq. 2, it provides an upper bound for it:

$$Pol_{inh}^{MC}(c) \geq Pol_{inh}(c) \forall c \tag{4}$$

4.2.3 Does it exist?

We compute the InPol for each comment across the four datasets. For Sap and Kumar, we use the exhaustive formulation (Eq. 2), whereas for the two DICES datasets we rely on the Monte Carlo approximation with $k = 1000$ (Eq. 3). Fig. 3 shows that in all DICES comments there exists at least one group that fully accounts for the observed polarization.[8] Interestingly, this observation does not hold for the Kumar and Sap datasets, despite evaluating all possible annotator groups exhaustively. See App. G for a qualitative look through comments with high InPol.

Finding 1. *Comments may exhibit a degree of InPol that cannot be attributed to any known or latent annotator grouping given the annotators present in the dataset.*

4.2.4 Why does it exist?

Hypothesis 1 (Low annotator count)**.** A low number of annotators per item may provide our model with insufficient information.

[7] Since polarization persists among male annotators, another subgroup of men may also contribute to it.

[8] $Pol_{inh}^{MC}(c) = 0 \Rightarrow Pol_{inh}(c) = 0$ due to Eq. 4. The same logic applies if we tried to increase the number of random partitions in Eq. 3.

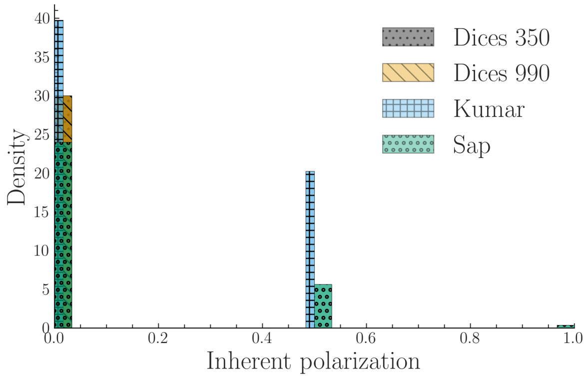

Figure 3: InPol across our datasets (§3). Contrary to assumptions made in prior work, InPol is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts.

Hypothesis 2 (Inaccessible groups)**.** InPol may arise from the presence of partially represented groups within the annotator pool. While these groups are responsible for the observed polarization, they may go undetected in our current methodology if they are represented by only one or two annotators, as these groups are discarded (§4.2).

Both of these explanations are plausible and reflect known limitations of most NLP datasets, although we lack such datasets with large annotator counts per item to provide a definitive answer. The DICES datasets can in principle be used for this purpose. However, even when repeating the experiment using random annotator subsets (ranging from 5 to 50 annotators) and taking the maximum across ten random samples for each subset size, *we still observed zero InPol* (App. H.1). This finding either invalidates both hypotheses, or indicates that the DICES datasets are structurally different. The latter explanation is plausible given that the annotation targets are LLM texts, which may be relatively unsubtle compared to human comments, as supported by low overall polarization (Fig. 6; App B).

4.3 Polarization is directional

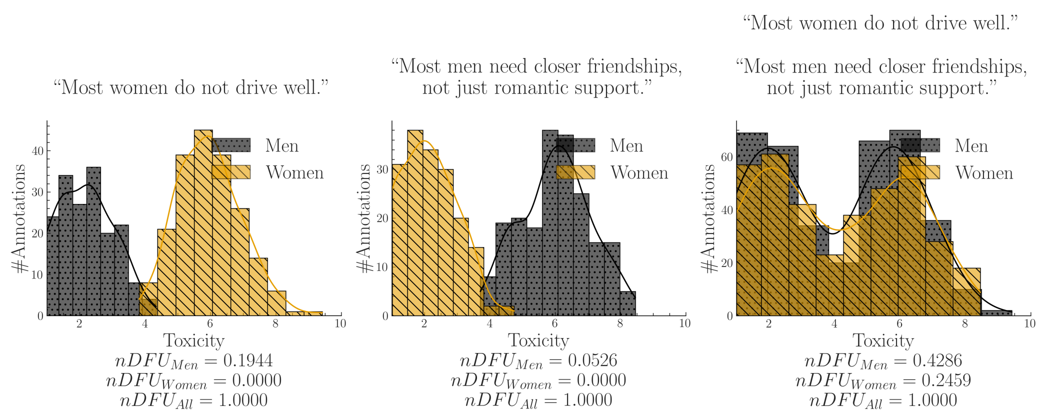

Figure 4: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization (polarization explained by gender, but not shown).

One way of potentially overcoming annotator sparsity, is aggregating annotations between comments. Fig. 4 shows an experiment where we combine annotations across two comments. We observe that, should the two comments be polarized for the opposite reason, the opposing polarization effects might balance each other out, leading to a false negative. Therefore, any statistics must be computed on the comment-level; annotations should never be aggregated across comments.[9]

Finding 2. *Polarization has a unique direction for each comment which may compromise cross-comment aggregation.*

5 How do we attribute the rest?

5.1 Apriori polarization

Instead of using InPol, which represents the minimum possible polarization exhibited by any group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*.

Definition 2 (Apriori polarization)**.** The expected polarization of a comment's annotations.

Specifically, we define apriori polarization for a comment c as:

$$Pol_{apr}(c) = \frac{1}{t}\sum_{i=1}^{t} \mathit{nDFU}(\tilde{r}_i(A(c))) \tag{5}$$

where $\tilde{r}$ is the random partition operator (see Eq. 2). We explain why defining apriori polarization in this way is preferable to simpler alternatives in App. C.1.

5.2 Expanding to the whole dataset

We can estimate how much polarization is generally present in the dataset by simply averaging the *apriori* polarization of its comments. Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any single group.

$$Pol_{apr}(d) = \frac{1}{|d|}\sum_{c \in d} Pol_{apr}(c) \tag{6}$$

We compare this quantity with the average observed polarization of all comments:

$$Pol_{obs}(d, g) = \frac{1}{|d|}\sum_{c \in d} \mathit{nDFU}(A(c \mid g)) \tag{7}$$

We can now compare the polarization that is attributed to a single annotator group (Eq. 7) to a prior one (Eq. 6). We define our normalized metric, *apunim*, as:

$$apunim(d, g) = \frac{Pol_{apr}(d) - Pol_{obs}(d, g)}{1 - Pol_{apr}(d)} \tag{8}$$

Definition 3 (Apunim)**.** The degree to which an annotator group contributes to overall polarization in a set of comments.

- $apunim(g) \approx 0$: The examined group does not meaningfully disagree with the other annotators.
- $apunim(g) > 0$: The examined group has fundamental disagreements with the other annotators.
- $apunim(g) < 0$: The annotators of that group are polarized between themselves.

[9] A similar assumption was followed in early parametric approaches, which assumed different parameters for each comment Checco et al. (2017).

5.3 Statistical significance

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student-t test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$\begin{aligned} H_0 &: \bar{apunim}_{\text{rand}}(d, g) = apunim(d, g) \\ H_a &: \bar{apunim}_{\text{rand}}(d, g) \neq apunim(d, g) \end{aligned} \tag{9}$$

where $\bar{apunim}_{\text{rand}}(d, g)$ denotes the average apunim values obtained from t randomized annotator partitions.

We account for the presence of multiple subgroups in each PC dimension using a multiple-hypothesis correction. For a detailed explanation of the p-value computation and the assumptions underlying the test, refer to App. C.3.

5.4 Does apunim recover a known effect?

Polarization and agreement metrics rest on different assumptions, which is why they are rarely compared directly. Fixing the size of a group effect by construction makes them commensurable, and we do so on synthetic annotations (App. I). Two outcomes bear on how §6 should be read. The *sign* of apunim carries information agreement-based attribution does not: a group that drives polarization and one that is merely noisy internally both depart from agreement, and Krippendorff's α scores them alike ($+.03$ vs. $+.02$), whereas apunim separates them ($+.32$ vs. $-.27$). Our significance test, however, is anti-conservative—it rejects at $\alpha = .05$ for 40–62% of samples containing no group effect—because the null is the *mean* of the reused random partitions (App. C.3), which narrows with the iteration count rather than with the evidence. A permutation p-value restores calibration.

Finding 3. *The sign of apunim separates between-group polarization from within-group disagreement, which agreement-based attribution conflates. Its p-values, however, are anti-conservative and should be recomputed by permutation.*

6 Attributing real-life polarization to human groups

6.1 Experimental setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can partially explain polarization in the annotations. Statistically significant results are presented in Table 2. See App. E for details on other parameters and execution times, and App. K for the full results.

6.2 Which factors contribute to polarization?

Ethnicity Polarization is consistently attributed to ethnicity across all datasets, confirming prior work suggesting that it is a systematic source of annotator disagreement in these tasks Goyal et al. (2022). We note that polarization attribution does not have a consistent direction across datasets, which we believe to be caused by studies using different ethnic subgroups, allowing polarization detection on the level of PCs, but distorting subgroup attribution.

Gender In both DICES-990 and Sap, women are a consistent source of polarization (positive values), while men disagree among themselves (negative values). These results are consistent with previous findings regarding HS (Wojatzki et al., 2018) and toxicity annotation Binns et al. (2017). Non-*transgender* annotators also showed the highest degree of polarization in the Kumar dataset in absolute terms ($-.788$) suggesting that transgender annotators are a systematic source of annotator disagreement, although only the inverse effect is detectable due to annotator sparsity (hence the negative sign).

Finding 4. *Annotator ethnicity and gender are consistent causes of polarization.*

Group	Apunim	Support
DICES-350		
Ethnicity=Asian	-.066	8138
Education=Other	.021	2191
DICES-990		
Ethnicity=AfricanAm	-.091	5810
Ethnicity=Latino	.133	6610
Ethnicity=Other	-.074	24321
Ethnicity=White	.120	11392
Age=GenZ	-.063	13741
Age=GenX+	.046	14384
Education=High school	-.075	7419
Gender=Man	-.027	32494
Gender=Woman	.024	33471
Sap		
Ethnicity=AfricanAm	.042	4
Gender=Man	-.091	1439
Gender=Woman	.244	877

Group	Apunim	Support
Kumar		
A. Innate Characteristics		
Ethnicity=White	-.029	6619
Education=College	-.034	2508
SexOrient=Heterosexual	-.050	5721
IsTrans=No	-.079	1697
B. Experiences and Ideologies		
HasBeenTargeted=False	-.048	6632
IsParent=No	-.069	5514
IsParent=Yes	.049	6216
ReligionImportant=No	-.062	3864
ReligionImportant=Very	.052	4062
SeenToxicity=False	.035	3050
SeenToxicity=True	-.077	6360

Table 2: Statistically significant ($p < 0.05$) apunim results across the four datasets. Gray rows show `Ethnicity` and `Gender` groups. "Support" refers to number of annotations. Consult App. K for the full results.

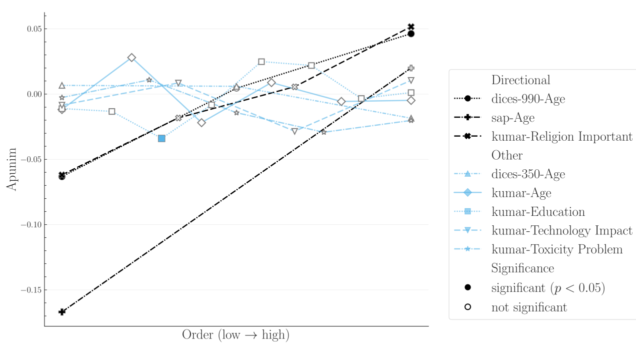

Figure 5: Polarization attribution trends in ordinal groups. The left side of the x-axis refers to young annotators, low education, low religiousness, and negativity towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in App. K. We exclude groups with < 50 annotations, since they near-universally result in zero apunim values due to lack of data.

Ideology and experiences appear to be a more consistent source of annotator disagreement than inherent characteristics. Specifically, the Kumar dataset, which tracks ideological attributes, shows polarization attribution linked primarily to personal experiences and beliefs, rather than to sociodemographic dimensions, although further datasets are required to generalize this finding. The divide between ideology/personal experiences and innate sociodemographic characteristics has not been adequately explored in current literature and should be further studied in future work.

Religion demonstrates a clear relationship with polarization attribution. According to Fig. 5, highly devout annotators exhibit higher levels of self-agreement compared with other annotators (positive apunim). Non-religious annotators show a pattern of increased internal disagreement (negative apunim). Notably, this trend of internal disagreement gradually reverses as religious devoutness increases (increasing apunim).[10]

[10]The statistical insignificance observed during the transition from irreligious to highly religious groups may represent genuine shifts rather than inconclusive results, since Eq. 9 checks whether $apunim \neq 0$, given that

Finding 5. *Pol. attribution may be more pronounced in ideological attributes and personal experiences. It also escalates as annotators become more religious.*

7 Conclusion

We investigated to what extent polarization can be attributed to annotator groups. We discovered as of yet unknown issues regarding the presence of inherent and conflicting polarization, which we treat using a new metric, *apunim*. By applying this metric to four subjective NLP datasets, we discover that (1) ethnicity and gender seem to generally drive polarization attribution, and (2) ideological groups may be more important than reported in the literature. Finally, we empirically investigate and discuss how our methodology interacts with (generally small in both annotations and size) current subjective NLP datasets (App. J) and release a PyPi library implementation to support further work.

Limitations

Data limitations Researchers often release only aggregated single-label annotations or omit annotator PC characteristics entirely. As a result, both our analysis and the applicability of our framework are limited by the scarcity of suitable datasets. In addition, because datasets with large numbers of annotators per item are rare, our approach operates only at the dataset level. This limitation prevents us from identifying individual comments or examples that could serve as concrete explanations for the observed *apunim* values.

Scope It must be noted that the datasets utilized in this study are restricted to posts written in English and pertain exclusively to the Western, specifically U.S., cultural context. Consequently, the findings and observed phenomena are not necessarily generalizable to linguistic or cultural contexts outside of this scope, such as non-English languages or other global cultural spheres.

Confounding variables Our analysis does not take into account confounding variables. It is plausible that some annotator groups are correlated with others (e.g., LGBTQ+ annotators may not lean into Conservatism). This is left for future work, as the scope of this analysis is deemed too large for this paper. Additionally, many datasets do not report anonymized IDs for the annotators, but only the annotations themselves, making this kind of analysis impossible. We leave our analysis as proof that our methodology can be used to surface polarization patterns, although future methods should allow for intergroup attribution.

Computational limitations Additionally, our metric features some important practical limitations. It is computationally expensive when applied to very large datasets with a very large amount of PC dimensions each featuring multiple distinct subgroups (App. E). It also tends to underestimate polarization attribution when applied to many tens of thousands of items (App. H). Furthermore, the p-value estimates may falsely give the impression that zero-attribution effects are genuine, which should be taken into account when interpreting results. These limitations do not render our method insufficient—the first two issues can be solved by subsampling. We nevertheless believe that NLP datasets should generally move away from huge annotated corpora, and instead towards smaller datasets with large annotator groups per item that sufficiently represent social subgroups.

Comparisons with other methods Finally, we compare against agreement-based attribution and against the original aposteriori unimodality test only on synthetic data, where the group effect is known by construction (App. I); on the datasets of §3 no ground truth exists against which to score them. That comparison also surfaced a calibration problem in our own significance test (Finding 3), which is the likeliest explanation for a pattern we could not previously account for—why some groups show statistically significant effects while others do not. The p-values in §6 should be read as descriptive until recomputed by permutation.

$\bar{apunim}_{\mathrm{rand}}(d, g)$ will usually be close to zero. This phenomenon may also apply to other ordinal PCs, such as `Age` in both Sap and DICES-990, but finer-grained subgroups are needed to verify this.

Ethics Statement

Dual use concerns The primary objective of this study is to identify systemic sources of polarization in subjective NLP datasets to facilitate the construction of more robust and equitable AI systems. However, given that our methodology allows for the quantitative attribution of specific judgment patterns to sociodemographic and ideological characteristics, we acknowledge three primary risks inherent in this research.

First, *Discriminatory Profiling*: If the metric is applied outside of a dataset quality analysis, its ability to correlate specific judgment patterns with annotator groups could be misused for surveillance or unjust profiling. Second, *Automated Bias Amplification*: The diagnostic patterns of polarization we detect are reflections of human systemic biases. Without human oversight, training subsequent models on these patterns risks automating and entrenching existing societal prejudices. Finally, the knowledge provided by *apunim* could be *weaponized*, allowing malicious actors to strategically target weak points in moderation systems or intentionally maximize societal conflict.

We stress that our metric functions strictly as a diagnostic warning signal. Any deployment in a real-world system must mandate a rigorous human-in-the-loop review, ensuring that identified patterns lead to systemic improvements (e.g., policy reform, demographic diversity initiatives) rather than merely justifying an automated system.

Data sourcing and privacy All datasets employed in this study were either publicly accessible or provided to us by the original authors with full disclosure regarding the purpose of our research. No identifiable, personal information was included in our pipeline. We do not release proprietary data provided by the authors.

Examples used Our study utilizes examples of comments that may be considered divisive across certain demographics. These examples were generated by our team solely for demonstration purposes. We explicitly state that the contents of these examples and the groups studied do not necessarily reflect the views or behaviors of the groups they represent, nor do they reflect the opinions of the authors.

Tools used We used small, open-source LLMs (primarily a local Gemma-4 model) to perform minor stylistic adjustments throughout this document. Furthermore, we utilized open-source and proprietary models to generate portions of the code for the plots and experiments presented. All such AI-generated outputs have been meticulously checked and validated by the authors.

Reproducibility Statement

All code, data-processing scripts and the library implementing *apunim* are released anonymously: library `https://anonymous.4open.science/r/apunim-5F70`, documentation `https://anonymous.4open.science/r/apunim-page`, replication code `https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`. Statistical procedures are detailed in App. C, replication details (seeds, compute) in App. E.

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Dataset	count	mean	SD	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8	1.17	69	72	73	74	76
Sap	585	5.6	.78	3	6	6	6	12
Kumar	2998	5.02	.34	5	5	5	5	10

Table 3: Number of annotations per item in each of the four datasets used in this study.

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A Acronyms

InPol Inherent Polarization

nDFU normalized Distance From Unimodality

NLP Natural Language Processing

GPLv3 GNU General Public Licence v3

AA African American

PC Personal Characteristic

HS Hate Speech

SD Standard Deviation

B Annotation details

The number of annotators per comment for all datasets presented in §3 can be found in 7, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We do not include these comments, since they are very likely a data error. Tables 4,5,6,7,8 show the annotation counts for each PC group in each dataset.

C Statistical considerations

C.1 Defining apriori polarization

In §4.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c \mid g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

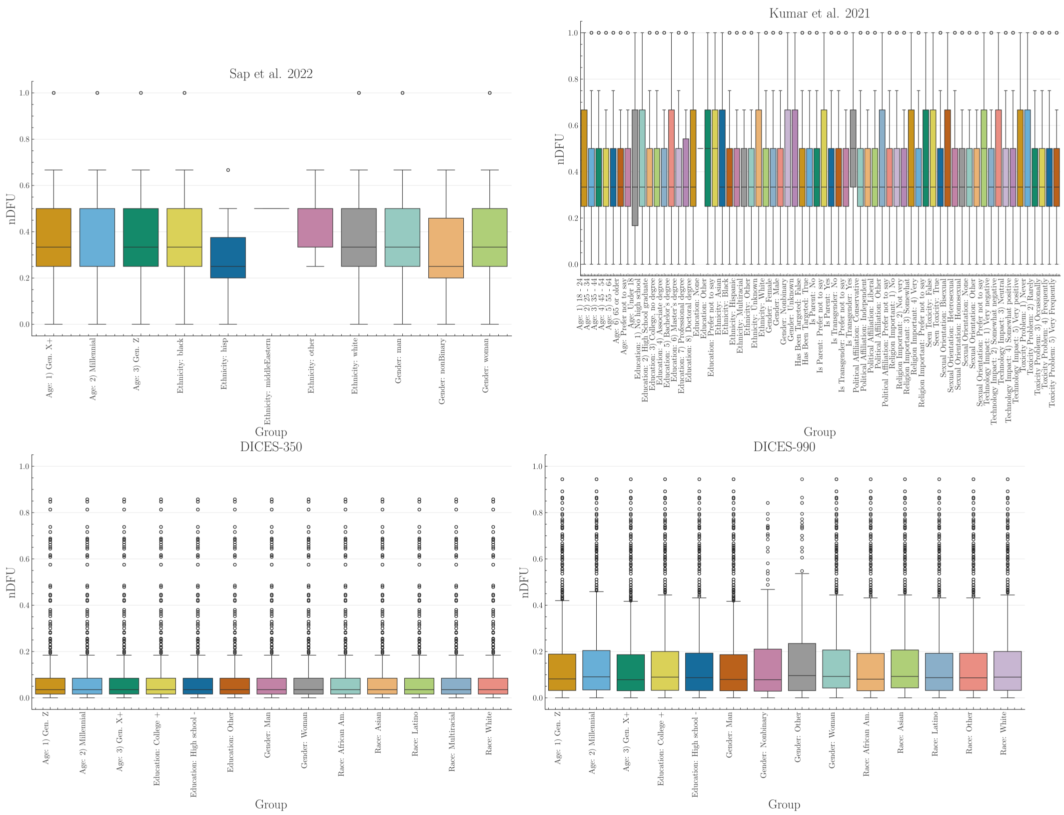

Figure 6: Boxplots of dataset polarization for the human datasets considered in this paper for all groups in each examined PC dimension.

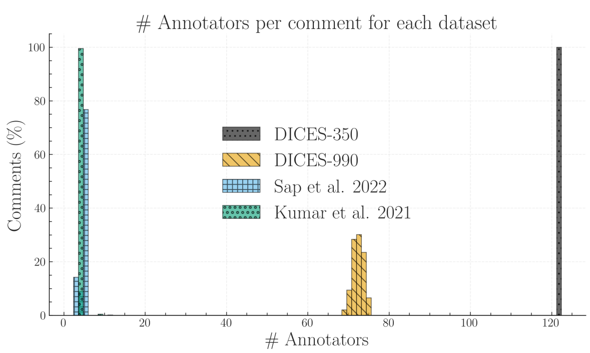

Figure 7: We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3), which is likely caused by a data error.

C.2 Filtering ineligible comments

Because *apunim* is fundamentally designed to quantify polarization, we exclude comments that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, we also exclude comments annotated exclusively by a single group (i.e., comments where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of nDFU (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only $5-6$ annotations per comment. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

Category	Subgroup	Support
Gender	Woman	21700
	Man	21350
Race	White	10500
	African Am.	10150
	Asian	9100
	Latino	7700
	Multiracial	5600
Age	1) Gen. Z	19600
	2) Millennial	12600
	3) Gen. X+	10850
Education	College +	26250
	High school -	14350
	Other	2450

Table 4: Subgroup Counts for DICES-350. “Support” refers to annotation count.

Category	Subgroup	Support
Gender	Woman	36050
	Man	35392
	Nonbinary	331
	Other	330
Race	Other	26394
	Asian	19885
	White	12322
	Latino	7170
	African Am.	6332
Age	Gen. Z	14899
	Millennial	41542
	Gen. X+	15662
Education	College +	64054
	High school -	8049

Table 5: Subgroup Counts for DICES-990. “Support” refers to annotation count.

C.3 P-VALUE ESTIMATION

Since our test is parametric (§5.3), it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per PC group to estimate nDFUs in Equations 5-7, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3). We discuss condition (2) in App. J. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method Holm (1979). The test is parameterized by the confidence level of our test, which

Category	Subgroup	Support
Age	Gen. Z	252
	Millennial	2259
	Gen. X+	786
Ethnicity	white	2168
	black	1091
	hisp	32
	other	5
	middleEastern	1
Gender	man	1859
	woman	1412
	nonBinary	26

Table 6: Subgroup Counts for Sap et al. 2022. “Support” refers to annotation count.

Category	Subgroup	Support
Gender	Female	8011
	Male	6832
	Unknown	124
	Nonbinary	93
Ethnicity	White	10699
	Black	1940
	Asian	862
	Multiracial	664
	Hispanic	418
	Other	247
	Unknown	230
Age	Under 18	3
	18 - 24	1711
	25 - 34	5969
	35 - 44	3785
	45 - 54	1977
	55 - 64	1101
	65 or older	471
	N/A	43
Education	No high school	71
	High School graduate	1353
	College, no degree	3090
	Associate degree	1676
	Bachelor’s degree	6071
	Master’s degree	2261
	Professional degree	236
	Doctoral degree	198
	N/A	71
	Other	32
Sexual Orient.	Heterosexual	12391
	Bisexual	1665
	Homosexual	515
	N/A	302
	Other	131

Table 7: Subgroup Counts for the Kumar et al. 2021 dataset. Part 1 of 2. “Support” refers to annotation count.

is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses Chen et al. (2017)).

Category	Subgroup	Support
Is Transgender	No	14549
	Yes	363
	N/A	148
Politics	Liberal	5997
	Independent	4143
	Conservative	3981
	N/A	609
	Other	330
Is Parent	Yes	7889
	No	7015
	N/A	156
Tech Impact	Very negative	149
	Somewhat negative	1411
	Neutral	1905
	Somewhat positive	7690
	Very positive	3905
Tox Problem	Never	869
	Rarely	2485
	Occasionally	5307
	Frequently	4441
	Very Frequently	1958
Rel Important	No	4735
	Not very	1808
	Somewhat	3548
	Very	4741
	N/A	228
Seen Toxicity	True	11505
	False	3555
Has Been Target	False	10741
	True	4319

Table 8: Subgroup Counts for Kumar et al. 2021 dataset. Part 2 of 2. “Support” refers to annotation count.

D Software

D.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (`https://anonymous.4open.science/r/apunim-page`).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project’s repository (`https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

D.2 Implementation details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{r}_i$, which are originally calculated to obtain the apriori polarization values (see Eq. 5), for the p-value calculations (§5.3). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{r}_i$ in Eq. 5), and the number of apunim values used in the p-value calculation have to be the same.

E REPLICATION DETAILS

We use version `1.0.6` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Eq. 5) and the p-value estimation (§5.3), and set a confidence level of $\alpha = 0.05$.

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of comments, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3 complete within a minute, the Kumar dataset necessitates approximately one hour of computation using commercial hardware. Furthermore, the annotator variance analysis detailed in App. J required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` package Tange (2025) for our own experiments). We also note that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered.

Finally, we note that in Fig. 10, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see App. B).

F SYNTHETIC EXPERIMENTS

This appendix details the synthetic experiments used to illustrate the functionality and behavior of nDFU. In all simulations, the bin size is set to $N_{bins} = 10$.

Disagreement vs. polarization To explore the relationship between annotation polarization and disagreement in Fig. 1, we constructed a 2×2 matrix of synthetic datasets, each drawing from a pool of $N_{annotators} = 100$. Polarization is simulated by defining a minority group, which constitutes 20% of the total annotators (n_special $= 20$). These minority annotators are assigned a targeted mean shift (mean_shift $= 1$) from the base distribution. The base variance for these distributions is set at variance $= 0.3$. Different types of distributions (unimodal, bimodal, multimodal, uniform) are achieved by controlling the base_means and applying variance multipliers (e.g., variance $\times\, 3$ for multimodal).

Single-comment polarization We simulate the distribution of toxicity annotations in Fig. 2 across two demographic groups (Men and Women). The distribution of Men's scores is modeled using a truncated normal distribution with loc $= 2$ and scale $= 1.3$, while Women's scores are modeled with loc $= 8$ and scale $= 1.3$. The combined dataset, representing all annotations, is used to demonstrate the impact of population weighting on the overall nDFU score. The sample size for each group in this visualization is $N_{annotators} = 50$.

Discussion-level polarization We examine nDFU scores for three specific textual examples presented in Fig. 4. For the individual comments, $N_{annotators} = 200$ for each group. In the first case (misogynist comment), Men's scores are modeled with loc $= 6$, while Women's scores are modeled with loc = 2. In the second case (misandrist comment), the roles are reversed. The third visualization combines distributions for men and women derived from the respective single-comment distributions.

G How does InPol look like?

Comment	InPol
The minks mentioned in the headline are indeed "rising" from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they're still very much dead, thankfully!.	1
There should be such thing as a bitch resistance for us women	1
There's a big difference between being a bitch and being a woman	1
i just watched a video of @joisnotonfire being an annoying hoe on skype i'm laughing	1
niggas love to commit crime	1
I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a nigga	0.5
I will become a cold ass bitch because of my treatment	0.5
I would happily lay out a nigga	0.5
Shoot yourself in the head you hypersensitive fuckin bitch.	0.5
not all women are bitches, but all bitches are women	0.5
" you ugly bitch!!!"- that seems to be my real name these days"	0
"I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!"" Omg, DED ;)"	0
"Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven."" Psalms 148:13 #Bible"	0
someone plz get this nigger some help. He's playing up like mad	0
stop talking you fucking nigger !	0

Table 9: Sample of comments from the Sap dataset and their InPol. Note that unambiguously toxic comments can be assigned with zero InPol, while more nuanced cases tend to have higher values.

Table 9 shows a random sample of comments taken from the Sap dataset and their respective InPol values (§4.2). Note that explicitly toxic or racist comments tend to have InPol values of zero, alongside clearly unproblematic comments. Inversely, more ambiguous and less comprehensible comments tend to have larger InPol values. This pattern may suggest that InPol may be tied to the overall ambiguity of the statements, especially in cases where comments may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of nDFU (§2.2). We include the InPol values for each individual comment for all four datasets in the supplementary material.

H Ablation experiments

H.1 Can DICES be used for InPol?

In order to investigate whether the numerous annotators per comments can explain the lack of InPol in the DICES datasets (§4.2.4) we gradually subsample the annotators for each comment randomly with replacement, starting from five annotators per comment, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the maximum InPol across all runs. Even with annotator counts similar to the Kumar and DICES datasets, InPol remains zero, suggesting that the absence of InPol can not be attributed to annotator number alone—rather, it is plausibly a pattern attributable to the DICES datasets themselves.

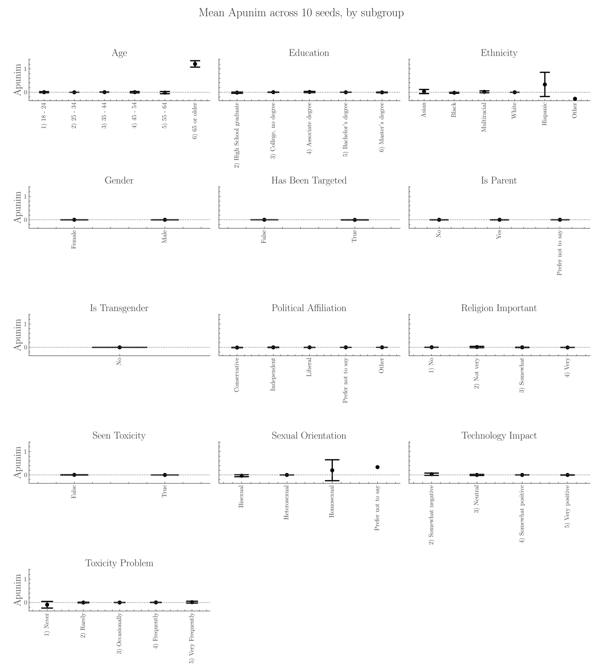

Figure 8: Apunim values on the Kumar dataset ($n_{samples} = 3000$) across ten runs with different subsets. Black bars indicate two SDs. PC dimensions with many groups are more volatile, although all dimensions broadly keep the same apunim sign across subsets.

H.2 How does dataset size influence apunim?

Since our approach was designed for the small-scale datasets common in NLP, we were cautious when testing it on the massive Kumar dataset. Because of the data volume, we analyzed various smaller samples (1,000, 10,000, and 30,000 comments), shown in Table 10 (rows with more than one N/A values removed). As sample size increased, the statistical metrics (apunim values) consistently shrank, and we found no statistically significant effects in the larger samples.

The test is overly conservative, meaning it almost always suggests there is no significant effect, even when an effect likely exists in reality. This inability to detect a real signal is a Type II Error (a false negative). This suggests that our test is statistically weak or "underpowered" in massive settings. Since our methodology is built for the small-scale NLP datasets that dominate the research field, we deem this limitation acceptable for the scope of our work.

H.3 How does sample section influence apunim?

Astute readers may notice that some variability does exist across differently sized samples for the same examined groups, where for example the sign of the apunim value may switch. We note that the statistically significant values (bold rows) change signs only when the apunim value approaches zero in the largest samples. Thus, the results are not inverted, but rather pushed towards zero.

In order to verify that this variability is caused by the sample size, and not by different samples being drawn, we execute a further ablation study. Specifically, we execute the apunim study on the 3000-sized sample of the dataset, using different subsets of the Kumar dataset. The results can be found in Fig. 8, where we note that variability is not an issue, except for a couple of cases. These cases seem to represent sparse annotator groups, where missing a few comments may genuinely change the value of the apunim metric. These groups however are largely not detected by apunim as significant either way (see §6).

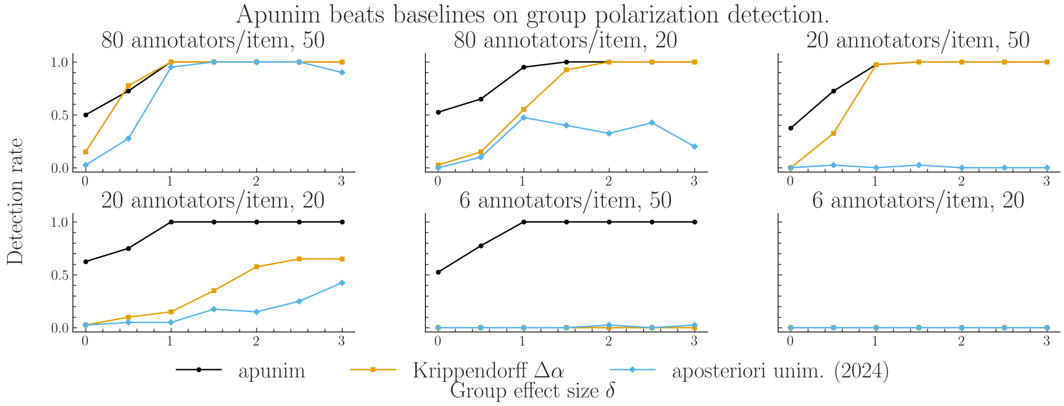

Figure 9: Detection rate for a group effect of known size δ; $\delta = 0$ is the false-positive rate and the dotted line marks the nominal $\alpha = .05$. Version `1.0.5` of our library binned each sample over its own range, which inflates the nDFU of any group that does not reach both ends of the scale; `1.0.6` bins every group and partition on one fixed grid.

I Comparison with agreement-based attribution

Setup We generate annotations on a 1–5 scale for 200 items, with the group effect fixed by construction: annotators in group B are shifted by δ relative to group A, so $\delta = 0$ means the grouping is irrelevant and larger δ means it explains more of the polarization. Every method receives identical data and is asked whether the grouping explains the polarization; we report the rate at which each rejects at $\alpha = .05$ over 40 replications. Annotations use a full crossed design (every annotator rates every item), which is the most favourable case for agreement metrics, as no cells are missing.

Baselines Krippendorff's α measures agreement rather than attribution, so the agreement baseline is the within-group gain it implies: split the annotators on the grouping and test whether α rises, calibrated by a label permutation test so that it carries a p-value on the same footing as apunim. We write this as $\Delta\alpha$. We also include the original aposteriori unimodality test Pavlopoulos & Likas (2024), which flags a comment that is polarized overall while every group is internally unimodal.

Power All methods recover a large effect (Fig. 9). They separate in the two regimes that matter for real corpora. With 20 annotators per item, the original test never fires at all ($\leq .05$ across every δ): it requires a group's nDFU to be exactly zero, and with few annotations per group that is a knife-edge condition, which is the concrete form of the noise problem raised in §2.3. With a 20% minority it recovers at most .40. Both apunim and $\Delta\alpha$ remain at ceiling in both regimes.

Specificity Power alone does not separate the methods, because a metric can fire for the wrong reason. Table 11 scores three cases where the grouping is not the explanation. When group B merely has higher variance around a shared mean—disagreement without bimodality—$\Delta\alpha$ fires every time, and cannot distinguish that case from genuine polarization (+.024 vs. +.031). Apunim also fires, but its value is strongly *negative* (−.268 vs. +.322), which is the reading given in §5.2: the group is polarized among itself. This signed separation is what a polarization metric buys over an agreement metric, and it is the main argument for preferring one.

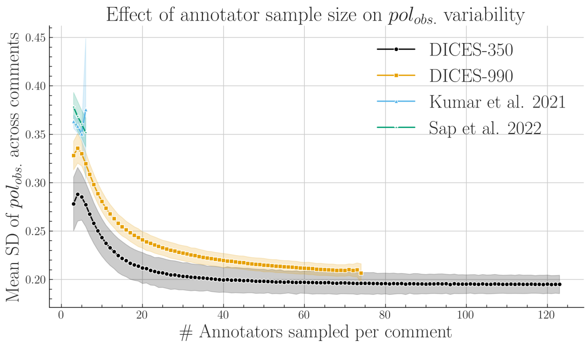

Figure 10: SD of pol_{obs} across comments, averaged across 1000 runs (Eq. 7). Shaded areas represent two SDs. Pol. estimates benefit from more annotators up to $n = 20$, at which point they begin to plateau.

	apunim	$\Delta\alpha$	orig. AU	should fire?
Group drives polarization	$+.322$	$+.031$	$+.058$	yes
Group is internally noisy only	$-.268$	$+.024$	$+.001$	by sign
Polarization unrelated to group	$+.002$	$+.000$	$+.006$	no
Real polarization, labels shuffled	$+.007$	$-.006$	$+.003$	no

Table 11: Mean statistic in four scenarios (80 annotators/item, balanced groups, $\delta = 2.5$ where applicable). Only apunim separates a group that drives polarization from one that is merely noisy, and it does so by sign.

Calibration The last two rows of Table 11 show apunim's statistic behaving correctly—it sits at zero when the grouping explains nothing—while its p-value does not: it rejects in 40% and 53% of those samples. The same is visible at $\delta = 0$ in Fig. 9, where the false-positive rate runs from .40 to .62 against a nominal .05, while $\Delta\alpha$ ($\leq$ .03) and the original test ($\leq$ .05) are calibrated. The cause is that the null is the mean of the reused random partitions (App. C.3); its standard error falls as $1/\sqrt{t}$ in the iteration count t, so significance can be obtained by computing longer. Holding the data fixed and raising t from 50 to 1600 leaves the rejection rate at $\approx$ .45 while the median p-value falls from .112 to .054. Scoring the same statistic against the permutation *distribution* rather than its mean gives a false-positive rate of .10 while retaining power (.93 at $\delta = 1$).

Effect of the binning fix Version `1.0.5` of our library passed a bin *count* to `numpy`, which spreads bins over the range of each sample. A group whose annotations do not reach both ends of the scale then acquires empty bins between populated ones, which DFU reads as multimodality. This is not symmetric: the size-matched random partitions mix all groups and so span the scale, while a concentrated group does not, which inflates Pol_{obs} relative to Pol_{apr} and drives Eq. 8 negative. On the data of Fig. 9 the statistic was negative in 88–100% of replications for $\delta \in [1.5, 2.5]$, where it must be positive. Binning every sample on one grid fixed over the annotation scale removes the inversion (+.143, +.222, +.322, +.428 for $\delta = 1.5$ to 3.0; negative in 0%).

J THE IDEAL SUBJECTIVE DATASET

We briefly discuss the type of data most suited for our proposed methodology, and connect it with current NLP datasets.

How many annotators? Our methodology does not directly require numerous annotators overall, only a sufficient number per item—indeed, statistically significant results can be obtained with as little as four annotations (see Sap-AA; Table 12). A large annotator pool however is required to get sufficient coverage on the examined groups. More annotators allow both higher fidelity of each PC dimension, and more such dimensions.

How many annotators per item? We investigate observed polarization against varying numbers of annotators. Specifically, for every comment, we repeatedly sampled n annotators with replacement, calculated pol_{obs} for each sample and measured SD, averaging it across 1000 same-sized samples. As shown in Fig. 10, pol_{obs} exhibits a clear, inverse relationship with the number of annotators, attributed to better histogram estimation provided by nDFU (§5.1), though the rate of gain substantially diminishes after 20 annotators. Interestingly, some variability persists even with a very high annotator count.

How many items? Our methodology is designed to work with smaller datasets common in NLP. Three of the four datasets in the study feature low (300–600) item counts, while the Kumar dataset has been subsampled to 3% of its original size. Large datasets should be used only to provide larger coverage on the target domain, since a much better representation of individual groups can be achieved with more annotators per item, rather than more items. Smaller datasets may also be more practical for social science analysis Rossi et al. (2024).

K FULL RESULTS

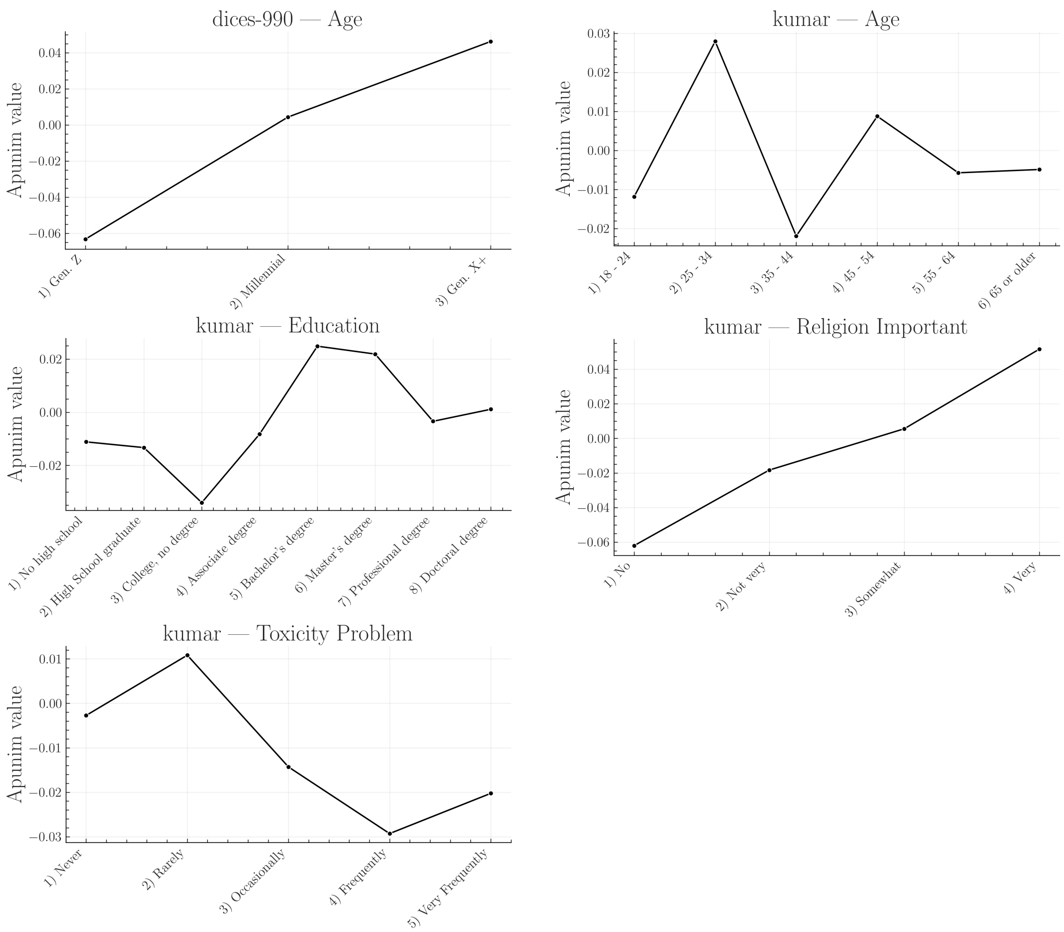

Figure 11: Apunim values expanded for each ordinal PC dimension. This figure is complementary to Fig. 5. The values in the graphs were extracted from the Tables of App. K.

PC	1000	10000	30000
Age			
8 - 24	-.002	-.037	-.016
25 - 34	.045	0	0
35 - 44	-.043	-.001	-.001
45 - 54	.001	.013	.003
55 - 64	.002	-.057	.001
65 or older	-.007	N/A	.948
Education			
High School graduate	-.023	-.005	.014
College, no degree	-.045	.010	.006
Associate degree	.001	.005	.009
Bachelor's degree	.029	-.001	-.000
Master's degree	.024	.004	-.000
Ethnicity			
Asian	-.012	.011	.007
Black	.036	-.004	-.012
Hispanic	.004	0	-.099
Multiracial	.008	0	0
Other	-.008	N/A	-.306
White	-.023	0	0
Gender			
Female	-.006	.002	.001
Male	.040	-.002	-.001
Has Been Targeted			
False	-.059	0	0
True	.017	-.003	-.002
Is Parent			
No	-.084	.001	.001
N/A	.004	1.000	0
Yes	.071	0	0
Is Transgender			
No	-.102	0	0
Political Affiliation			
Conservative	.056	-.003	-.002
Independent	-.003	.001	.002
Liberal	-.026	.001	.001
Other	-.003	0	0
Religion Important			
No	-.071	.002	.001
Not very	-.013	.010	.006
Somewhat	.022	.003	-.002
Very	.043	-.003	-.002
Seen Toxicity			
False	.025	-.001	-.001
True	-.082	.001	0
Sexual Orientation			
Bisexual	-.003	-.040	-.013
Heterosexual	-.067	0	0
Homosexual	-.012	0	.053
Technology Impact			
Somewhat negative	-.008	0	.003
Neutral	.017	-.014	0
Somewhat positive	-.053	0	.001
Very positive	.027	-.004	-.003
Toxicity Problem			
Never	-.008	-.037	-.081
Rarely	.004	.003	-.001
Occasionally	-.012	-.003	-.001
Frequently	-.032	-.001	0
Very Frequently	-.017	.030	.015

Table 10: Comparison of apunim values across different sample sizes (1k, 10k, 30k) for the Kumar dataset. Bold rows show statistically significant results in the 3k sample used in the main text.

Group	apunim	pvalue	support
Kumar			
Age=18-24	-.0118	.4822	1431
Age=25-34	.0279	.1185	4888
Age=35-44	-.0219	.1733	3102
Age=45-54	.0088	.5394	1668
Age=55-64	-.0057	.6592	938
Age=65 or older	-.0049	.6747	409
Age=N/A	.0000	N/A	37
Age=Under 18	.0000	N/A	2
Ed=No HS	-.0111	.5278	60
Ed=High School	-.0133	.3461	1123
Ed=College	-.0340	.0189	2508
Ed=Associate	-.0082	.5963	1362
Ed=Bachelor's	.0249	.1091	5007
Ed=Master's	.0219	.1091	1982
Ed=Professional	-.0034	.7459	192
Ed=Doctoral	.0011	.8651	164
Ed=Other	.0000	N/A	28
Ed=N/A	.0000	N/A	63
Eth=Asian	-.0019	.9332	720
Eth=Black	.0143	.7070	1671
Eth=Hispanic	-.0075	.9092	352
Eth=Multiracial	-.0026	.9332	540
Eth=Other	-.0007	.9332	204
Eth=Unknown	-.0063	.9092	189
Eth=White	-.0291	.0416	6619
Gen=Female	-.0245	.0720	6141
Gen=Male	.0261	.0720	5479
Gen=Nonbinary	-.0014	.9436	81
Gen=Unknown	-.0008	.9436	104
Targeted=False	-.0482	1.04e-05	6632
Targeted=True	.0146	.2310	3643
IsParent=No	-.0686	2.39e-08	5514
IsParent=N/A	.0017	.8913	130
IsParent=Yes	.0490	2.85e-05	6216
IsTrans=No	-.0788	7.47e-05	1697
IsTrans=N/A	-.0007	.9483	128
IsTrans=Yes	-.0073	.7711	340
Pol=Conservative	.0269	.0534	3386
Pol=Independent	-.0201	.1371	3432
Pol=Liberal	-.0278	.0534	4945
Pol=Other	-.0132	.2947	271
Pol=N/A	.0034	.6523	511
RelImp=No	-.0620	2.96e-07	3864
RelImp=Not very	-.0183	.1039	1493
Ed=Somewhat	.0056	.6169	3012
RelImp=Very	.0516	4.58e-05	4062
RelImp=N/A	.0066	.2968	194
SeenTox=False	.0350	.0029	3050
SeenTox=True	-.0766	4.82e-13	6360
SexOrient=Bisexual	.0087	.7884	1481
SexOrient=Heterosexual	-.0500	3.22e-05	5721
SexOrient=Homosexual	-.0048	.7884	430
SexOrient=Other	.0009	.9055	115
SexOrient=N/A	-.0061	.7884	265
TechImp=Very neg	.0000	N/A	118
TechImp=Somewhat neg	-.0085	.4097	1179
TechImp=Neutral	.0085	.4097	1603
TechImp=Somewhat	-.0285	.0691	6040
TechImp=Very pos	.0105	.4097	3315
ToxProblem=Never	-.0027	.8071	741
ToxProblem=Rarely	.0108	.3965	2087
ToxProblem=Occ	-.0143	.3809	4347
ToxProblem=Freq	-.0293	.0507	3740
ToxProblem=Very	-.0202	.1404	1625

Table 12: Apunim results for the Kumar dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value.

Group	apunim	pvalue	support
Sap			
Age=GenX+	-.066	.708	271
Age=Millennial	.007	.744	1886
Ethnicity=black	.042	4.000	4
Ethnicity=white	-.000	.998	1896
Gender=Man	-.091	.000	1439
Gender=Woman	.244	.000	877

Table 13: Apunim results for the Sap Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

Group	apunim	pvalue	support
DICES-350			
Age Gen. Z	.007	.539	17528
Age Millennial	.006	.539	11268
Age Gen. X+	-.019	.166	9703
Ed=College +	-.012	.234	23475
Ed=High school	-.015	.186	12833
Ed=Other	.021	.025	2191
Gender Man	.019	.058	19093
Gender Woman	-.021	.058	19406
Race African Am.	.043	2.41e-05	9077
Race Asian	-.066	1.16e-10	8138
Race Latino	.003	.920	6886
Race Multiracial	.000	.980	5008
Race White	.020	.067	9390

Table 14: Apunim results for the DICES-350 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

Group	apunim	pvalue	support
DICES-990			
Age Gen. Z	-.063	3.05e-25	13741
Age Millennial	.004	.495	38436
Age Gen. X+	.046	7.17e-15	14384
Ed=College +	-.009	.163	59142
Ed=High school	-.075	5.80e-42	7419
Gender Man	-.027	3.81e-05	32494
Gender Woman	.025	.00015	33471
Race African Am.	-.091	1.61e-64	5810
Race Asian	-.011	.080	18428
Race Latino	.133	7.57e-112	6610
Race Other	-.075	1.51e-30	24321
Race White	.120	6.03e-81	11392

Table 15: Apunim results for the DICES-990 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.